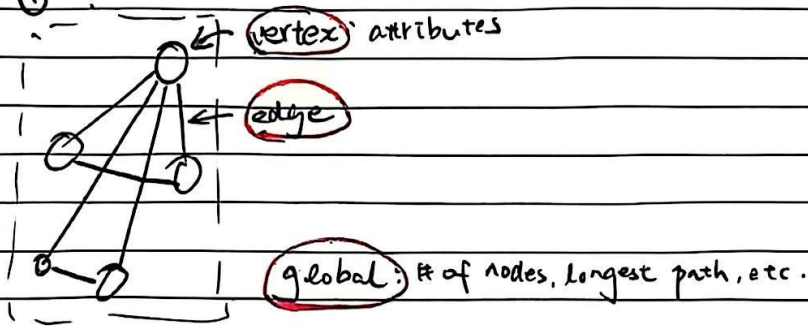


Graph Neural Network.

- parts:
- 1> what kind of data is most naturally phrased as graph
 - 2> how graphs differs from other types of data.
specialised choices we have to make
 - 3> build 'GNN'

Part ①:



○ — ○ Undirected Edge

○ → ○ Directed Edge

Where types of problems have graph structured data?

3 general types of predictions:

1. graph-level
2. node-level
3. edge-level.

node-level: using an analogous to image segmentation, we trying to label the role of each pixel.

Entities to make prediction: nodes, edges, global context, and connectivity

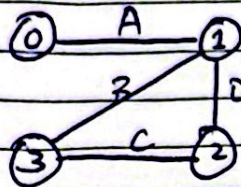
3 Ways to represent the graph:

1> ~~Adjacency~~ Adjacency Matrix.

2> Adjacency List

3> Incidence Matrix.

exp 1:



(undirected edges).

① Adjacency matrix

		To			
		0	1	2	3
from	0		1		
	1	1		1	1
	2	0	1		1
	3		1	1	

(Symmetry, sparse).

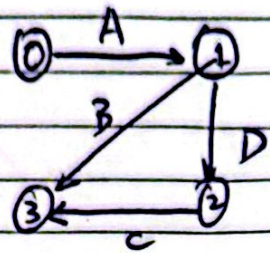
② Adjacency List.

0	[1]
1	[0, 2, 3]
2	[1, 3]
3	[1, 2]

③ Incidence Matrix.

	A	B	C	D
0	1	1	0	0
1	1	0	0	1
2	0	0	1	1
3	0	1	1	0

exp 2:



(directed edges).

① Adjacency Matrix:

	0	1	2	3
0	0	1	0	0
1	0	0	1	1
2	0	0	0	1
3	0	0	0	0

② Adjacency List:

0	[1]
1	[2, 3]
2	[3]
3	[]

③ Incidence Matrix

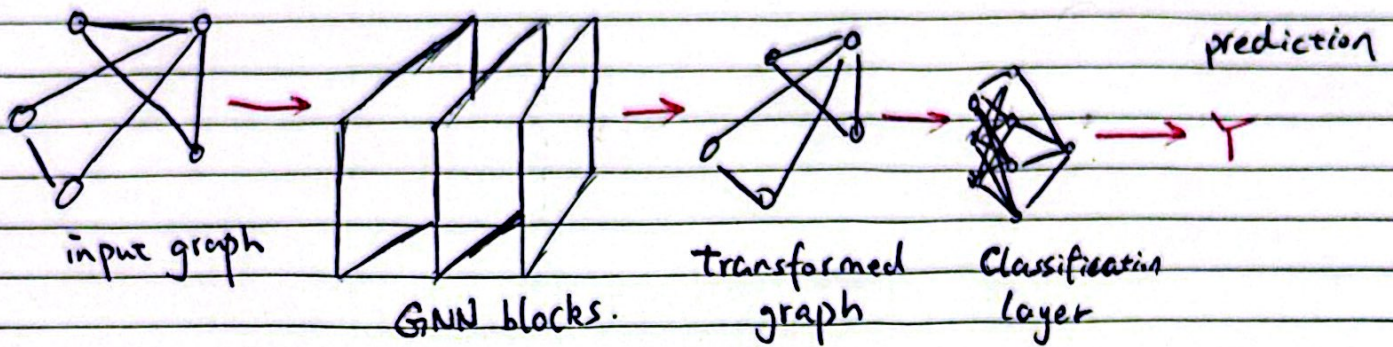
	A	B	C	D
0	1	0	0	0
1	-1	1	0	1
2	0	0	1	-1
3	0	-1	-1	0

NB: → Adjacency Matrix & Incidence Matrix are permutation variant, whereas adjacency list is not.

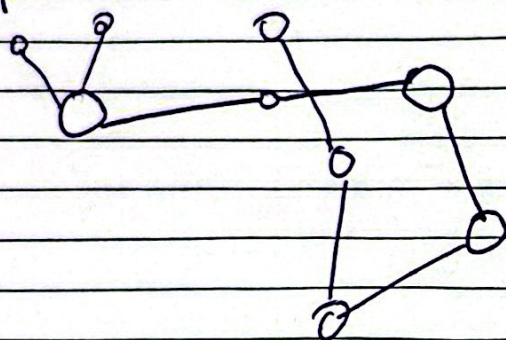
→ Adjacency list is ^{less} sparse compare to Adjacency matrix.

3) Adjacency list can be represented by a tensor.

Example of using GNN model to performing end-to-end prediction:



Example with chemical molecule:



graph level task: pungent

node level task

Edge level task

parameters:

- ① Depth

- ② Aggregation function

- ③ Node Embedding Size

- ④ Edge Embedding Size

- ⑤ Global Embedding Size.